

Online E-Commerce Website Using Agile Methodology

1. Abstract

Online shopping has become very popular due to the growth of the internet and digital technology. An Online E-Commerce Website allows customers to buy products online without visiting physical stores. Users can view products, place orders, and make payments easily from anywhere.

This project focuses on developing an **Online E-Commerce Website using Agile methodology**. The system provides features such as user registration, product browsing, cart management, online payment, and order tracking. All customer and order details are stored securely in a centralized database.

Agile methodology is used to develop the system in small phases called sprints. Each sprint includes planning, design, development, testing, and review. Agile helps in faster development and allows changes based on customer feedback. This makes the system flexible and efficient.

The proposed system reduces manual work and improves customer satisfaction. It is user-friendly, secure, and reliable. In the future, features like mobile app support, recommendation systems, and advanced analytics can be added.

2. Introduction

2.1 Introduction

An E-Commerce website is an online platform where customers can buy and sell products over the internet.

2.2 Problem Identification

- Manual shopping is time-consuming
- Limited product availability
- No easy order tracking

2.3 Need of the Project

- 24×7 online shopping
- Easy product comparison
- Secure payment system

2.4 Project Scheduling

The project is divided into multiple sprints using Agile methodology.

2.5 Objectives

- Provide online product shopping
- Manage users, products, and orders
- Improve customer experience

3. Software Requirement Specification

3.1 Purpose

To develop an online e-commerce system for managing products and customer orders.

3.2 Scope

Used by customers, sellers, and administrators.

3.3 Hardware / Software Requirements

Hardware

- Computer / Laptop
- Minimum 4 GB RAM

Software

- OS: Windows / Linux
- Language: Java / Python
- Database: MySQL

3.4 Tools

- VS Code / Eclipse
- Web Browser
- Database Server

3.5 Software Process Model

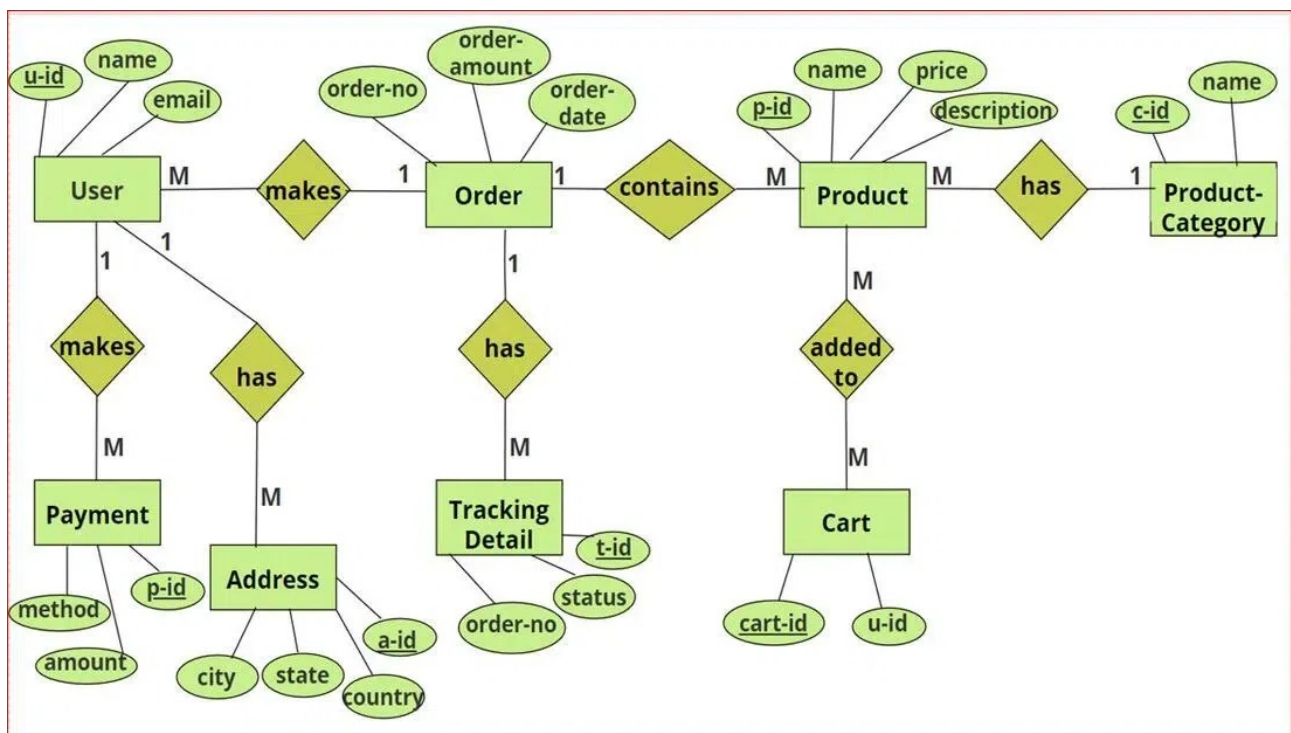
Agile Software Development Model.

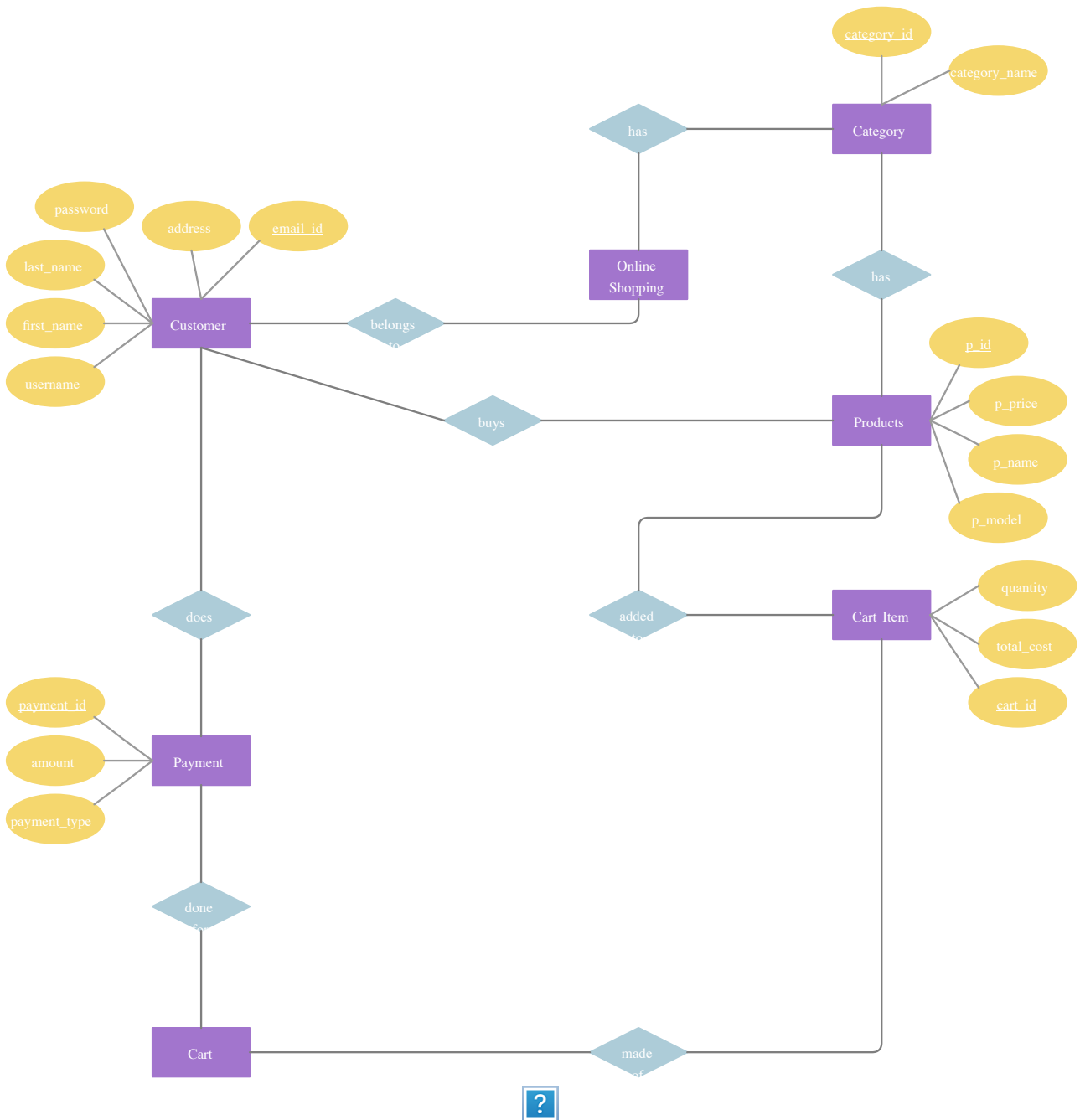
4. System Design

4.1 Data Dictionary

Field Name	Description
UserID	Unique user ID
ProductID	Unique product ID
ProductName	Name of product
OrderID	Order number
PaymentStatus	Payment details

4.2 ER Diagram (REAL)

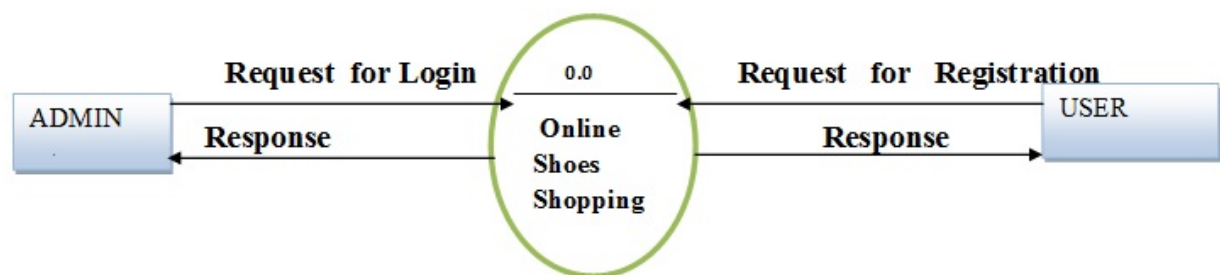
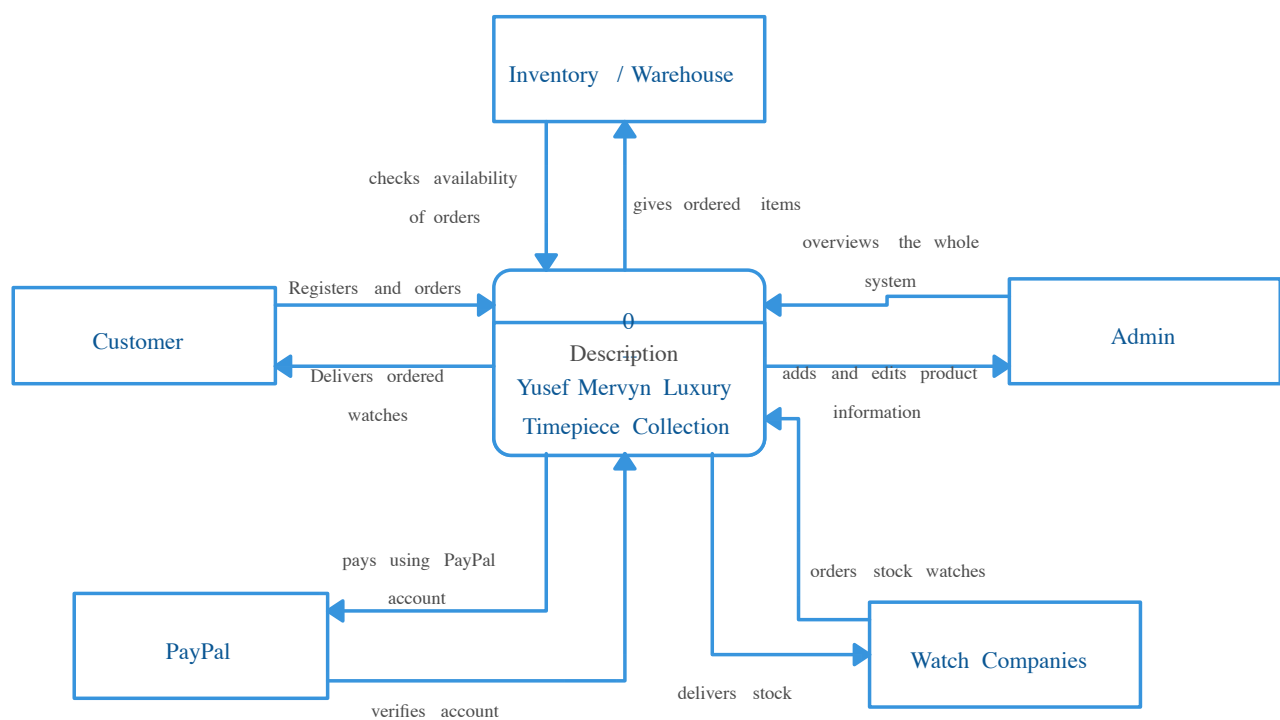




Entities to draw:

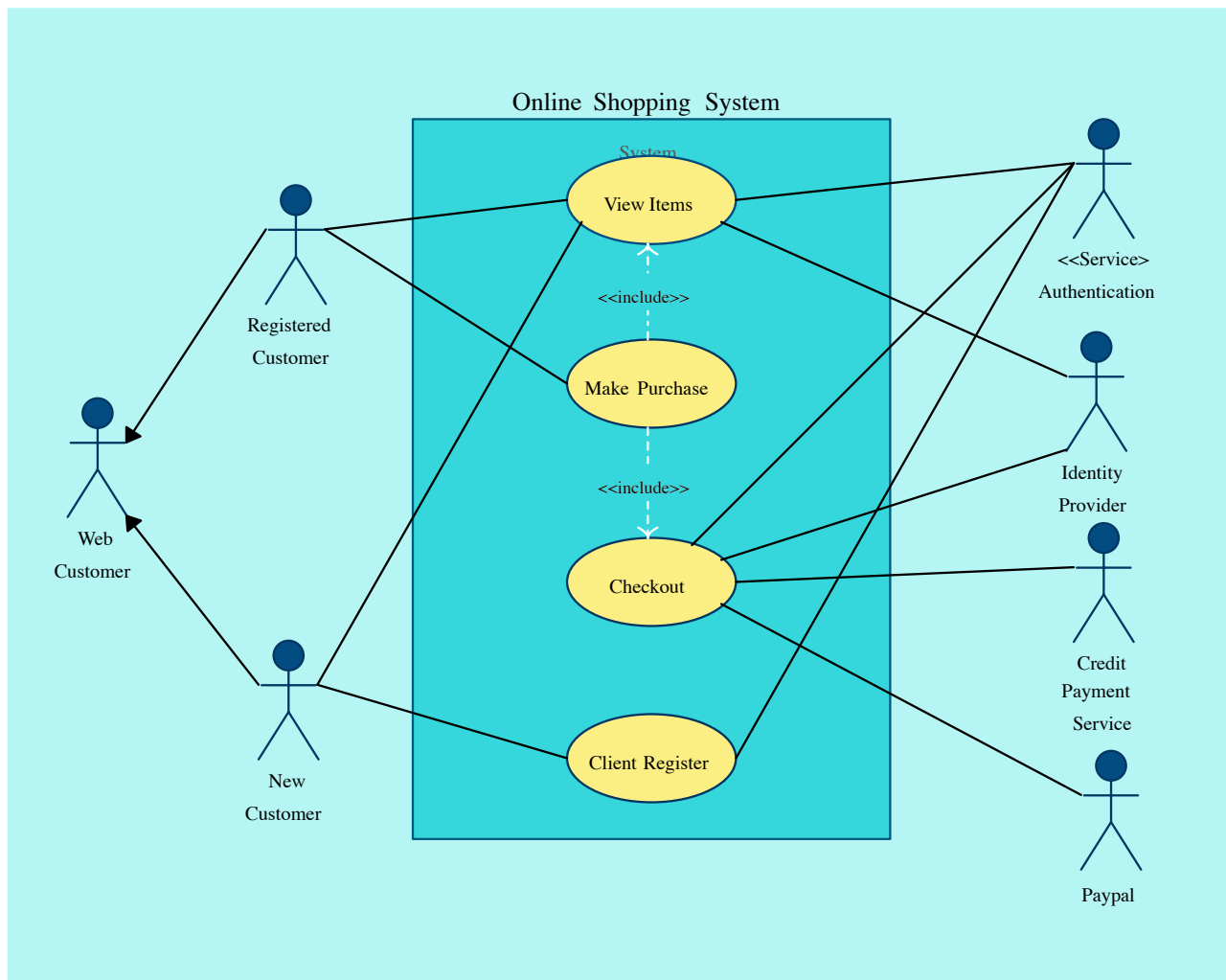
- USER (UserID, Name, Email)
- PRODUCT (ProductID, Name, Price)
- ORDER (OrderID, Date, Status)
- PAYMENT (PaymentID, Amount, Status)

4.3 DFD – Level 0



Flow:
User → E-Commerce System → Order Confirmation
System ↔ Database

4.4 Use Case Diagram



Use Cases:

- Register / Login
- Browse Products
- Add to Cart
- Place Order
- Make Payment

4.5 System Flow Chart

Flow:

Start → Login → Browse Products → Add to Cart → Payment → Order Confirmed → End

5. Implementation

5.1 Program Code

The e-commerce system is implemented using Java/Python with MySQL database.

5.2 Output Screens

- Login Page
- Product Listing Page
- Order Confirmation Page

6. Testing

6.1 Test Data

- Valid user and order data
- Invalid login details

6.2 Test Result

All modules work correctly and give expected output.

7. User Manual

7.1 How to Use

1. Register and login
2. Browse products
3. Add items to cart
4. Make payment
5. View order details

7.2 Screen Layout

Simple and user-friendly interface.

8. Project Applications and Limitations

Applications

- Online shopping websites
- Retail businesses
- E-commerce companies

Limitations

- Internet connection required
- Payment gateway dependency

9. Conclusion and Future Enhancement

The Online E-Commerce Website provides a convenient way for customers to shop online. Agile methodology ensures flexible and fast development. Future enhancements include mobile apps, product recommendations, and AI-based search.

10. Bibliography & References

- Software Engineering Textbook
- Agile Methodology Notes
- Online E-Commerce Resources